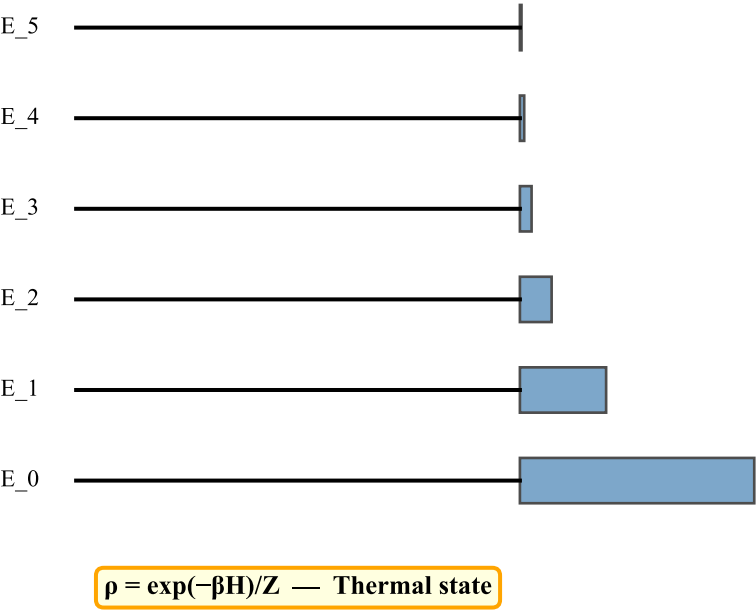


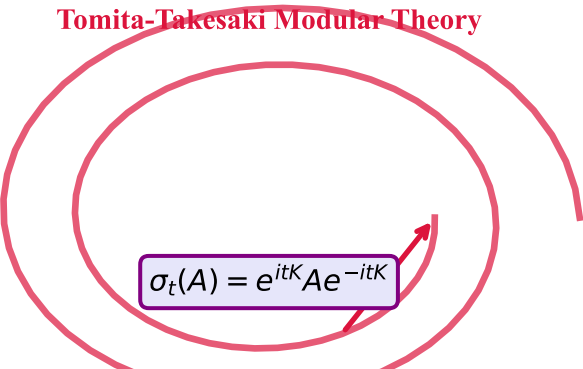
Thermal Time Hypothesis: Time Emerges from Statistical State

Statistical Mechanical System



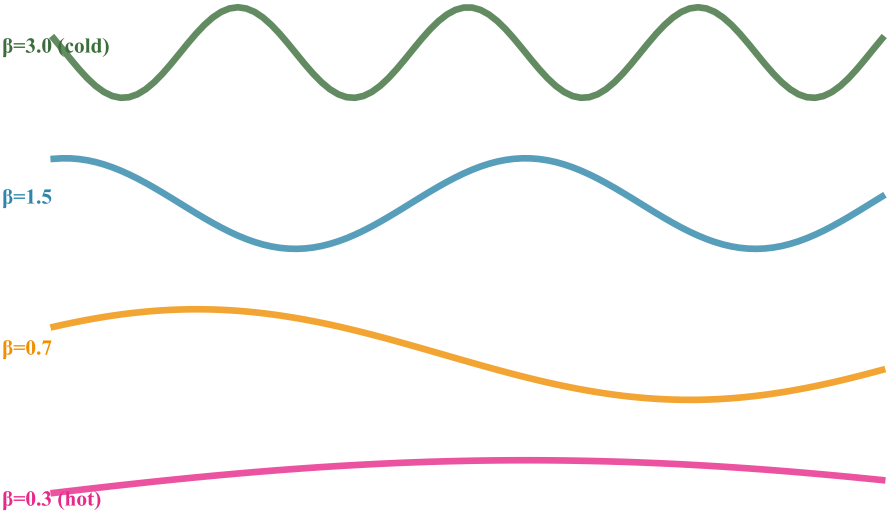
Modular Hamiltonian: $K = -\ln(\rho)$

$$K = -\ln(\rho) = \beta H + \ln(Z)$$

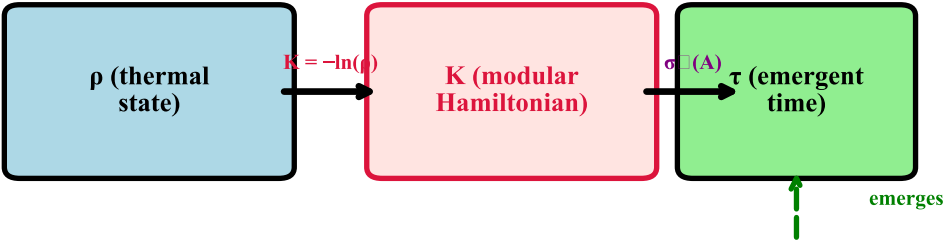


Different $\beta \rightarrow$ Different Time Flows

$$\tau = \beta \times (\text{thermodynamic time})$$



Thermal Time: From State to Flow



Key Insight:

*There is no absolute time flow —
time evolution is determined by the state ρ itself*

→ Time is not a background structure

Time is not fundamental — it emerges from thermal state